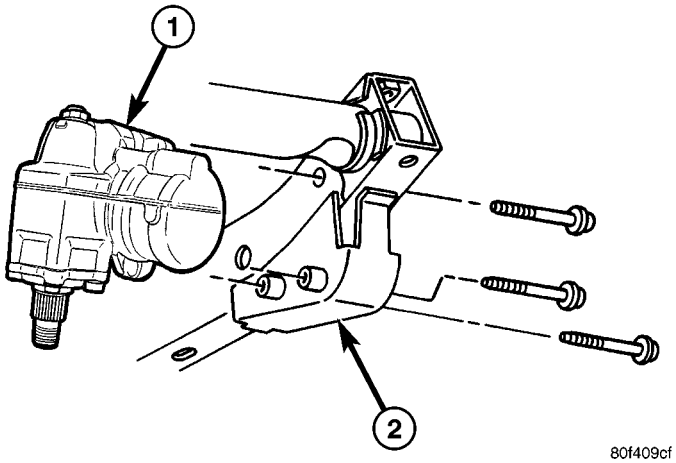

Fig. 5 STEERING GEAR REMOVAL/INSTALL

1 - STEERING GEAR
2 - FRAME MOUNT

INSTALLATION

(1) Install steering gear on the frame rail and tighten bolts to 95 N·m (70 ft. lbs.)

(2) Align and install the pitman arm and tighten nut to 251 N·m (185 ft. lbs.).

(3) Align the column coupler shaft to steering gear. Install a **new** coupler pinch bolt and tighten to 49 N·m (36 ft. lbs.).

(4) Install power steering hoses/tubes to steering gear and tighten to 28 N·m (21 ft. lbs.).

(5) Install the bumper shield.

(6) Fill power steering system to proper level, (Refer to 19 - STEERING/PUMP - STANDARD PROCEDURE).

ADJUSTMENTS

CAUTION: Steering gear must be adjusted in the proper order. If adjustments are not performed in order, gear damage and improper steering response may result.

NOTE: Adjusting the steering gear in the vehicle is not recommended. Remove gear from the vehicle and drain the fluid. Then mount gear in a vise to perform adjustments.

WORM THRUST BEARING PRELOAD

NOTE: Off center torque (180 to 360 degrees from gear center) = 0.4 to 0.8 Nm (3.5 in-lb to 7.0 in-lb) This torque is set in the assembly plant and can't be adjusted in the field

OVER - CENTER ROTATING TORQUE

(1) Mount the gear carefully into a vise.

CAUTION: Do not overtighten the vise on the gear case. This may affect the adjustment

(2) Rotate the input shaft with a crows foot socket from stop to stop and count the number of turns.

(3) Starting at either stop, turn the input shaft back 1/2 the total number of turns. This is the center of the gear travel.

(4) Loosen the adjuster plug locknut.

(5) Place the torque wrench in the vertical position on the input shaft. Rotate the wrench 50 degrees each side of the center and record the highest rotational torque in this range (Fig. 6). This is the Over-Center Rotating Torque.

NOTE: The input shaft must rotate smoothly without sticking or binding.

(6) The Over-Center Rotating Torque should be 0.53-0.93 N·m (4.5 - 8 in. lbs.) **higher** in addition to the off center torque from above (Fig. 6). than the Preload Rotating Torque.

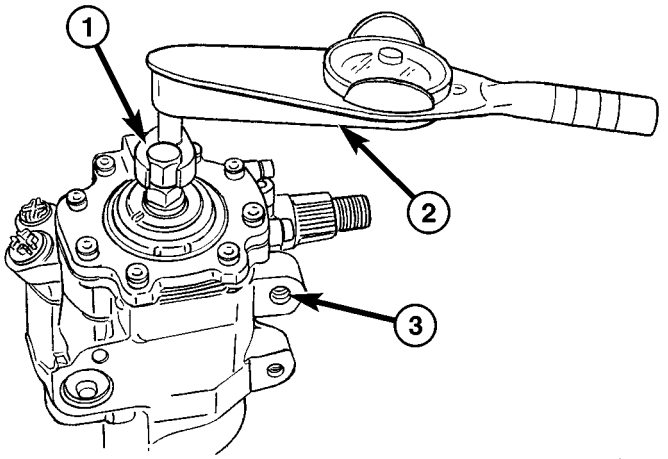

Fig. 6 OVER-CENTER TORQUE

1 - CROWSFOOT WRENCH
2 - INCH POUND TORQUE WRENCH
3 - STEERING GEAR

(7) If an adjustment to the Over-Center Rotating Torque is necessary, first loosen the adjuster lock nut. Then turn the pitman shaft adjuster screw back (COUNTERCLOCKWISE) increases torque, (CLOCKWISE) reduces torque (Fig. 7).

(8) Remeasure Over-Center Rotating Torque. If necessary turn the adjuster screw and repeat measurement until correct Over-Center Rotating Torque is reached (Fig. 7).

NOTE: To increase the Over-Center Rotating Torque turn the screw COUNTERCLOCKWISE.

(9) Prevent the adjuster screw from turning while tightening adjuster lock nut (Fig. 7). Tighten the adjuster lock nut to 65 N·m (48 ft. lbs.).